Chickens, Eggs, and Granger Causality

Min Kyeong Baik

University of Illinois at Urbana Champaign

October 24, 2016

Outline

- Replication of Thurman and Fishers Table 1 (Granger Causality Test)
- Suggestion of Graphical Technique (Partial Residual Plot)
- Augmented Dickey-Fuller Test (Unit Root Test)
- Re-estimate the Granger Causality Model in Difference

Data

- Annual time series data
  - U.S. egg production (in millions of dozens)
  - December 1st USDA estimate of U.S. chicken population excluding broilers
- Sample priod : 1930 - 2004 (75 years)

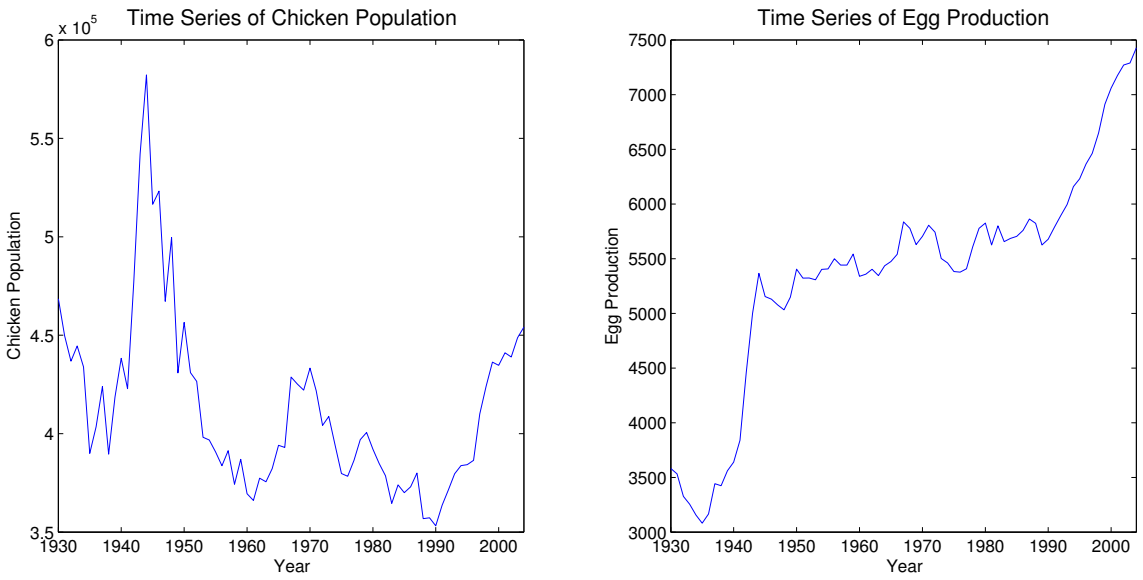


Figure : Time Series Plot of Chicken Population and Egg Production

Granger Causality Test

- If lagged values of X help predict current values of Y in a forecast formed from lagged values of both X and Y, then X is said to Granger cause Y. (Thurman and Fisher (1988))
- In order to conclude either eggs or chickens come first to the other, we need to have a result in an one way direction. If we have either both cause each other or both don't cause each other, we can not say further about the Granger causality between the two variables.
- The Granger causality test is basically the F-test for joint significance of the OLS coefficients.

Granger Causality Test

- We consider the following two regression equations. For each equations, we first do OLS estimation and do joint F-test for the coefficient β_i's. The null hypothesis is $H_0 : \beta_1 = \cdots = \beta_L = 0$, and the alternative hypothesis is $H_1 : \beta_i \neq 0$, for at least one i. We use one to four lags, i.e., $L = 1, \cdots, 4$.
- Test if chickens Granger cause eggs

$$Eggs_t = \mu + \sum_{i=1}^{L} \alpha_i Eggs_{t-i} + \sum_{i=1}^{L} \beta_i Chickens_{t-i} + \varepsilon_t$$

  If the coefficient β_i's are jointly significant, we can say that chickens Granger cause eggs.
- Test if eggs Granger cause chickens

$$Chickens_t = \mu + \sum_{i=1}^{L} \alpha_i Chickens_{t-i} + \sum_{i=1}^{L} \beta_i Eggs_{t-i} + \varepsilon_t$$

  If the coefficient β_i's are jointly significant, we can say that eggs Granger cause chickens.

Granger Causality Test

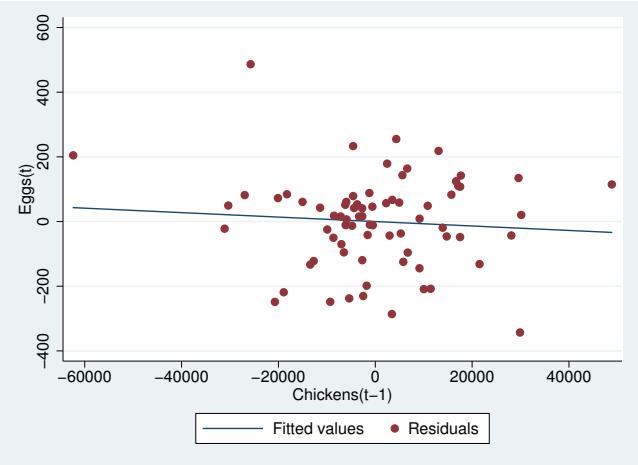
Part 1 : Did the Chicken Come First?			
$H_0 : \beta_1 = \cdots = \beta_L = 0$ (Chickens do not Granger cause eggs)			
No. of Lags	F-statistic	P-value	R^2 of the Regression
1	0.01	0.9321	0.9763
2	0.64	0.5297	0.9795
3	0.47	0.7072	0.9791
4	0.32	0.8641	0.9776

Part 2 : Did the Egg Come First?			
$H_0 : \beta_1 = \cdots = \beta_L = 0$ (Eggs do not Granger cause chickens)			
No. of Lags	F-statistic	P-value	R^2 of the Regression
1	0.08	0.7846	0.7633
2	9.58	0.0002	0.8158
3	5.46	0.0021	0.8240
4	4.47	0.0031	0.8292

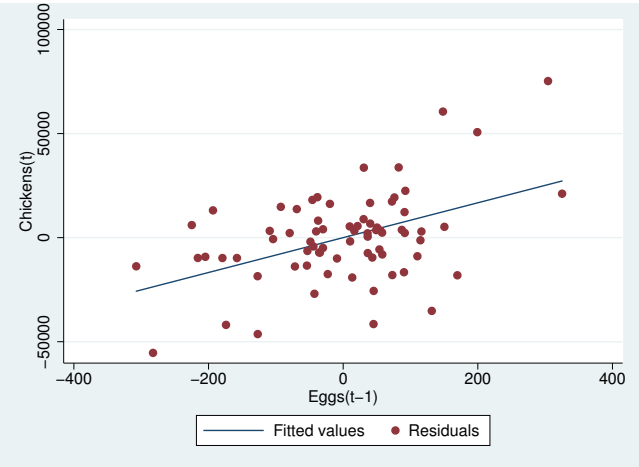
Table : Granger Causality Test (1930-2004, Annual Data)

- We can conclude that eggs Granger cause chickens, while chickens do not Granger cause eggs. Thus eggs come first to chickens, as Thurman and Fisher (1988) derived.

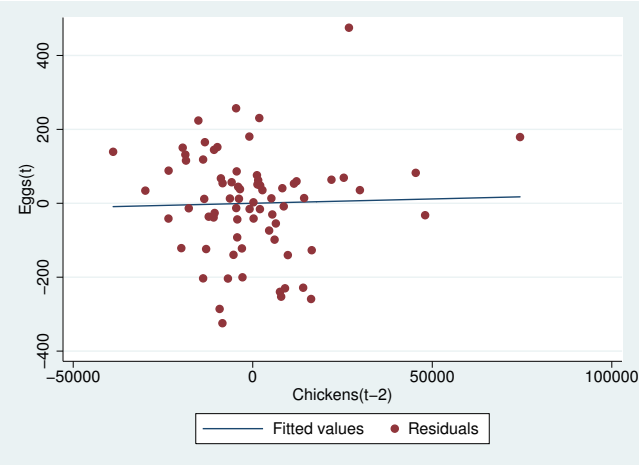
Partial Residual Plot



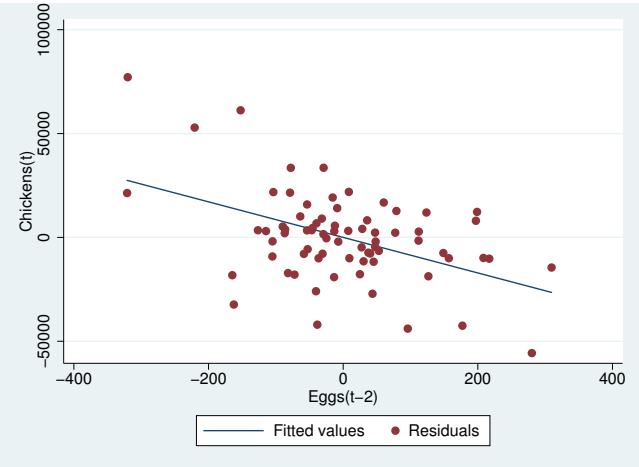
(a) $Eggs_t$ against $Chickens_{t-1}$



(b) $Chickens_t$ against $Eggs_{t-1}$



(c) $Eggs_t$ against $Chickens_{t-2}$



(d) $Chickens_t$ against $Eggs_{t-2}$

Partial Residual Plot

- Comparing the slope of fitted lines and the range of axis, we can see that plots for eggs against lagged chickens have a flat fitted lines, while plots for chickens against lagged eggs have relatively steeper fitted lines.
- By these partial residual plot analysis, we can conclude that the lagged values of chicken population does not have a significant effect on the current egg production, while the lagged values of egg production have a significant effect on the current chicken population.

Augmented Dickey-Fuller Test

- The Augmented Dickey-Fuller test is an augmented version of the Dickey-Fuller test for a larger and more complicated set of time-series models.
- Consider a simple AR(1) model

$$y_t = \rho y_{t-1} + u_t$$

- Under a complicated error structure of u_t (ex. ARMA(1,1)), we consider a model

$$\triangle y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{L} \delta_i \triangle y_{t-i} + \varepsilon_t,$$

  where $\gamma = \rho - 1$, $\varepsilon_t \sim i.i.d.$
- The null and alternative hypothesis for the Augmented Dickey-Fuller test is $H_0 : \gamma = 0$ (a variable has a unit root) against $H_1 : \gamma < 0$ (the variable follows a stationary process)

Augmented Dickey-Fuller Test

- Three versions of the ADF test
  - Case 1 : Random walk w/o drift and trend
    $$\triangle y_t = \gamma y_{t-1} + \sum_{i=1}^{L} \delta_i \triangle y_{t-i} + \varepsilon_t$$
  - Case 2 : Random walk with drift and w/o trend
    $$\triangle y_t = \alpha + \gamma y_{t-1} + \sum_{i=1}^{L} \delta_i \triangle y_{t-i} + \varepsilon_t$$
  - Case 3 : Random walk with drift and trend
    $$\triangle y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{L} \delta_i \triangle y_{t-i} + \varepsilon_t$$
- The t-statistic does not follow t-distribution. So in order to determine whether to reject or not reject the null hypothesis, we use the Table B.6 in Hamilton (1994). We can also use the critical values that are interpolated based on the tables in Fuller (1996).
- We conclude that both time series of eggs and chickens are random walk (i.e., nonstationary). Relating two unit root series with each other does not yields a valid causal relationships. Thus the results of the Granger causality test in the previous part can be derived from a "spurious regression".

Augmented Dickey-Fuller Test

No. of Lags	*Case 1: Without Both Drift and Trend*				
	Test Statistic		Critical Value		
	Eggs	Chickens	1%	5%	10%
1	1.719	-0.185	-2.611	-1.950	-1.610
2	1.830	-0.100	-2.611	-1.950	-1.610
3	1.844	-0.155	-2.612	-1.950	-1.610
4	1.912	-0.111	-2.612	-1.950	-1.610

No. of Lags	*Case 2: With Drift and Without Trend*				
	Test Statistic		Critical Value		
	Eggs	Chickens	1%	5%	10%
1	-0.809	-1.908	-3.548	-2.912	-2.591
2	-1.167	-2.289	-3.549	-2.912	-2.591
3	-1.205	-2.319	-3.551	-2.913	-2.592
4	-1.423	-2.673	-3.552	-2.914	-2.592

No. of Lags	*Case 3: With Both Drift and Trend*				
	Test Statistic		Critical Value		
	Eggs	Chickens	1%	5%	10%
1	-1.988	-1.678	-4.099	-3.477	-3.166
2	-2.186	-2.281	-4.102	-3.478	-3.167
3	-2.196	-2.268	-4.104	-3.479	-3.167
4	-2.359	-2.843	-4.106	-3.480	-3.168

Table : Results of the ADF Test on Eggs and Chickens

- We can not reject the null hypothesis under 5% significance level, since there is no test statistic which is less than 5% critical value.

Granger Causality Test for Models in Difference

- Since the unit root hypotheses are plausible, we re-estimate the Granger causality models in differences.
- Similar to the previous part, we consider the following two regression equations, but here, we use the difference and lagged difference terms in the model. For each equations, we first do OLS estimation and do F-test for the coefficient β_i's. The null hypothesis is $H_0 : \beta_1 = \cdots = \beta_L = 0$, and the alternative hypothesis is $H_1 : \beta_i \neq 0$, for at least one i. We use one to four lags, i.e., $L = 1, \cdots, 4$.
- Test if changes in chickens Granger cause changes in eggs

$$\triangle Eggs_t = \mu + \sum_{i=1}^{L} \alpha_i \triangle Eggs_{t-i} + \sum_{i=1}^{L} \beta_i \triangle Chickens_{t-i} + \varepsilon_t$$

- Test is changes in eggs Granger cause changes in chickens

$$\triangle Chickens_t = \mu + \sum_{i=1}^{L} \alpha_i \triangle Chickens_{t-i} + \sum_{i=1}^{L} \beta_i \triangle Eggs_{t-i} + \varepsilon_t$$

Granger Causality Test for Models in Difference

Part 1 : Did the Chicken Come First?			
$H_0 : \beta_1 = \cdots = \beta_L = 0$ (Chickens do not Granger cause eggs)			
No. of Lags	F-statistic	P-value	R^2 of the Regression
1	0.12	0.7294	0.1478
2	0.25	0.7817	0.1479
3	0.15	0.9318	0.1413
4	0.20	0.9377	0.1432

Part 2 : Did the Egg Come First?			
$H_0 : \beta_1 = \cdots = \beta_L = 0$ (Eggs do not Granger cause chickens)			
No. of Lags	F-statistic	P-value	R^2 of the Regression
1	12.38	0.0008	0.1700
2	4.42	0.0157	0.1755
3	3.36	0.0239	0.1962
4	4.16	0.0048	0.2792

Table : Granger Causality Test in Differences (1930-2004, Annual Data)

- We can conclude that changes in eggs Granger cause changes in chickens, while changes in chickens do not Granger cause changes in eggs. Thus eggs come first to chickens, same as the result we obtain in previous part.